Polynomial Identities

Name _____

Algebra 2 Lesson 3-3 (fast track) Date _____ Period _____

Learning goal: I can use Pascal's Triangle and the Binomial Theorem to expand a power of a binomial.

Language goal: I can explain how I expanded a binomial using the words row, coefficient, exponent, and term.

On the quiz: (1) Name the reason for each step of an identity proof (full Savvas quiz only). (2) Bonus: multiply $(5-4x^3)(5+4x^3)$ and name the identity (reference box, page 3). (3) Factor $1-125n^3$ (full Savvas quiz only). (4) Expand $(x+2)^8$: Row 8 with $a=x$, $b=2$, like Example 5A. (5) Fill in the coefficients of $(a+4)^5$: Row 5 times the powers of 4, like Example 5B.

On the Topic 3 test: Question 3 asks you to expand $(x-2)^6$ with the Binomial Theorem, and Question 13 asks which terms appear in $(2a+2b)^5$ (Example 5). Question 8 (factor $64+27a^3$) comes from Example 3, which this packet skips.

Pascal's Triangle: Add the two numbers diagonally above

Row 0	1
Row 1	1 1
Row 2	1 2 1
Row 3	1 3 3 1
Row 4	**1 4 6 4 1**
Row 5	1 5 10 10 5 1
Row 6	___ ___ ___ ___ ___ ___ ___
Row 7	___ ___ ___ ___ ___ ___ ___ ___
Row 8	___ ___ ___ ___ ___ ___ ___ ___ ___

Fill in Rows 6, 7 and 8. If there is no second number above, think of it as 0, so every row starts and ends with 1.

Binomial Theorem. For every positive integer n,

$$(a+b)^n = C_0a^n + C_1a^{n-1}b + \cdots + C_nb^n$$

The coefficients $C_0, \ldots, C_n$ are the numbers in **Row** n. The powers of a go down, the powers of b go up, and the exponents in each **term** add to n.

Example 5: Apply the Binomial Theorem

Use the Binomial Theorem to expand the expressions.

A. Find $(x-3)^4$.

1. The exponent is 4, so $n=4$:
$C_0a^4 + C_1a^3b + C_2a^2b^2 + C_3ab^3 + C_4b^4$

2. Row 4 (red) gives the coefficients:
$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

3. Identify a and b: $a=x$ and $b=-3$ (subtracting, so use the opposite of 3).

4. Substitute, then simplify:
$x^4 + 4x^3(-3) + 6x^2(-3)^2 + 4x(-3)^3 + (-3)^4$
$= x^4 + 4x^3(-3) + 6x^2(9) + 4x(-27) + 81$
$= x^4 - 12x^3 + 54x^2 - 108x + 81$

B. Find $(s^2+3)^5$.

1. Row 5: $1, 5, 10, 10, 5, 1$, so
$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$

2. $a=s^2$ and $b=3$.

3. Substitute:
$(s^2)^5 + 5(s^2)^4(3) + 10(s^2)^3(3)^2$
$+10(s^2)^2(3)^3 + 5(s^2)(3)^4 + (3)^5$

4. Simplify (multiply exponents of s; $3^2, \ldots, 3^5$ are $9, 27, 81, 243$):
$= s^{10} + 15s^8 + 90s^6 + 270s^4 + 405s^2 + 243$

Check: $(a+b)^n$ has $n+1$ terms (5 in A, 6 in B). Signs alternate in A: odd powers of -3 are negative.

Say it: To expand $(x-3)^4$ I used Row _____ because the exponent is _____. The coefficients are _____. In each term the exponent of x goes _____ and the exponents add to _____.

Try It 5: Use the Binomial Theorem to expand each expression

Hint: Use your Rows 6 and 7. Write the pattern with a and b first, then substitute.

a. $(x-1)^7$

Row 7: _____ $a =$ _____ $b =$ _____

b. $(2c+d)^6$

Row 6: _____ $a =$ _____ $b =$ _____

Say it: In part b, why is the first term not $2c^6$? Use the words *coefficient* and *exponent*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
A7M ★	Pascal's triangle and the Binomial Theorem	Quiz 4 and 5: pick the row, write the coefficients
NEU ★	Binomial Theorem II	Quiz 4 and 5: substitute a and b (a number, maybe negative), simplify each term
G7Y	Pascal's triangle	Optional warm-up: build Rows 6–8 (A7M covers the rows the quiz needs)
NJV	Factor sums and differences of cubes	Optional: Quiz 3 (full Savvas quiz only), Topic 3 test
8GN	Multiply polynomials	Warm-up from 3-2 for Bonus Quiz 2

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Reference: the polynomial identities (Concept Summary)

Difference of Squares: $a^2 - b^2 = (a+b)(a-b)$	Square of a Sum: $(a+b)^2 = a^2 + 2ab + b^2$
Difference of Cubes: $a^3 - b^3 = (a-b)(a^2+ab+b^2)$	Sum of Cubes: $a^3 + b^3 = (a+b)(a^2-ab+b^2)$

Exit question: Expand $(x+i)^4$ and justify your work

Use Pascal's Triangle and the Binomial Theorem to expand $(x+i)^4$. Justify your work.

Row 4: _____ $a =$ _____ $b =$ _____ $i^2 =$ ____ $i^3 =$ ____ $i^4 =$ ____

Before the quiz, finish these sentences.

To expand $(a+b)^8$ I use Row _____ of Pascal's Triangle, and the expansion has _____ terms.

Each _____ in the expansion comes from the row; the _____ of a goes down by 1 in each _____.

When the binomial is $(x-3)$, I use $b =$ _____, so the signs of the terms _____.